

④ (iv) When $n = 4.5$
 $y = 5.8 - 0.514 \times 4.5 = 3.487$

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

⑤ Same as ④

⑥ in ④

Chap-7

Probability Distribution:

① Binomial distribution $\rightarrow n < 20$
 ② Poisson $\rightarrow$ Discrete r.v.

③ Normal $\rightarrow$ continuous r.v.

① What is Binomial distribution?

$\Rightarrow$ A discrete random variable X is said to have Binomial distribution if its Probability function is as follows

$$f(x) = {}^n C_x p^x q^{n-x} ; x = 0, 1, 2, \dots, n$$

Where, n & p are the parameters and

$$p + q = 1$$

Hence, n = number of trial.

p = Probability of success.

q = () of failure.

Problem: A family has 5 children. The probability of being a boy and a girl is equal. Find the prob probability that the family has

- ① No boys.
- ② Exactly 2 boys.
- ③ At most 2 u.
- ④ At least 3 u.

Hence

$$n = 5.$$

$$p = q = \frac{1}{2} \quad \left(\begin{array}{l} \text{the probability of being a boy or} \\ \text{a girl is equal} \end{array} \right)$$

Therefore,

$$f(x) = {}^n C_x p^x q^{n-x}; \quad x = 0, 1, 2, \dots, n$$
$$= {}^5 C_x \left(\frac{1}{2}\right)^x \cdot \left(\frac{1}{2}\right)^{5-x}; \quad x = 0, 1, 2, \dots, 5$$

$$\textcircled{1} f(x=0) = {}^5 C_0 \left(\frac{1}{2}\right)^0 \cdot \left(\frac{1}{2}\right)^{5-0} = \frac{1}{2^5}$$

$$(41) P(X=2) = {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2}$$

$$(42) P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$(43) P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$$

Ch-7

Extra

→ Binomial Math

→ 1, 2, 3 Math practice

Poisson

Poisson distribution

$$P(X) = \frac{e^{-m} \cdot m^x}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

Where, m is the parameter and $e = 2.718$

Note:- Mean of Binomial distribution

$$= np$$

Mean of Poisson distribution = m

Practical example of Poisson distribution:

Stat

Poisson distribution:

$$P(x) = \frac{e^{-m} \cdot m^x}{x!}; x = 0, 1, 2, \dots, \infty$$

Problem:- If 2% of electric bulbs are ~~manufactured~~ manufactured by a certain company are defective. Find the probability that in a sample of 200 bulbs -

- ① Exactly 2 bulbs.
- ② Less than 2 u.
- ③ More than 3 u are defective.

Note:- mean of binomial

Soln:-

We know,

The probability function of a poisson distribution

$$P(x) = \frac{e^{-m} \cdot m^x}{x!}; x = 0, 1, 2, \dots, \infty$$

Now, $m = np$.

$$= 200 \times \frac{2}{100}$$

$$= 4.$$

$$\text{Therefore, } P(x) = \frac{e^{-4} \cdot 4^x}{x!}; x = 0, 1, 2, \dots, \infty$$

~~Q1 Q2~~

$$1) f(x=2) = \frac{e^{-4 \cdot 4^2}}{2!}$$

$$2) f(x < 2) = f(x=0) + f(x=1)$$

$$3) f(x > 3) = \frac{f(x=0) + f(x=1) + f(x=2) + f(x=3) + f(x=4) + f(x=5) + \dots + f(x=\infty)}{=1}$$

$$\rightarrow = 1 - [f(x < 4)]$$

⑥ Normal distribution:

⇒ A continuous random variable 'X' is said to have normal distribution (if its Probability density func is as follows)

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

$-\infty < x < \infty$
 $-\infty < \mu < \infty$
 $\sigma^2 \geq 0$

Where μ & σ are the parameters.
— mean
— standard deviation

Here,

⑦ Importance of Normal distribution is
 computer Science. (5 sentence → from net)